**AD 502 – Machine Learning**

**Pre-Reading Material – Lecture 38**

**Basic Ensemble Techniques: Out of Bag Evaluation**

# Pre-Reading Material – Out-of-Bag (OOB) Evaluation

Estimated Reading Time: 5 Minutes

## 1. Introduction

Out-of-Bag (OOB) Evaluation is a model evaluation technique commonly associated with Bagging and Random Forest algorithms.

In Bagging, multiple bootstrap samples are generated from the original training dataset using sampling with replacement. Since sampling is performed with replacement, some observations may appear multiple times in a bootstrap sample, while others may not be selected at all.

The observations that are not selected for training a particular model are called Out-of-Bag (OOB) samples.

OOB Evaluation uses these unused observations to estimate the performance of the trained model without requiring a separate validation dataset.

## 2. Working Principle of OOB Evaluation

OOB Evaluation follows these steps:

1. Start with the original training dataset.
2. Generate multiple bootstrap samples using sampling with replacement.
3. Train a separate machine learning model on each bootstrap sample.
4. Identify the observations that were not selected for each bootstrap sample.
5. Use these unused observations to test the corresponding trained model.
6. Collect predictions from different models for each observation.
7. Combine the predictions and calculate the overall evaluation performance.


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**3. Understanding Out-of-Bag Samples**

Consider a dataset containing 100 observations.

When a bootstrap sample is generated, some observations may be selected multiple times, while others may not be selected.

The observations that are not included in a particular bootstrap sample are called OOB samples for that model.

Approximately 36.8% of the original observations remain unused in each bootstrap sample, assuming a bootstrap sample has the same size as the original dataset.

These unused observations can be used for model evaluation.

**4. OOB Evaluation in Random Forest**

Random Forest uses multiple Decision Trees trained on different bootstrap samples.

For a particular training observation:

- Some Decision Trees may have used that observation during training.
- Other Decision Trees may not have used it.
- The trees that did not use the observation can provide predictions for that observation.
- These predictions are combined to obtain its OOB prediction.

For classification, majority voting is generally used, whereas for regression, predictions are averaged.

The OOB predictions are compared with the actual target values to estimate model performance.

**5. Advantages of OOB Evaluation**

1. No Separate Validation Dataset: It provides an alternative to maintaining a separate validation dataset.
2. Efficient Evaluation: Model performance can be estimated during the training process.
3. Better Data Utilization: It allows observations to contribute to training some models and evaluation of others.


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4. Overfitting Assessment: It helps estimate how well a model may perform on unseen observations.
5. Convenient for Random Forest: It provides an inbuilt evaluation mechanism in many Random Forest implementations.

**6. Limitations of OOB Evaluation**

1. It is mainly applicable to ensemble methods based on bootstrap sampling.
2. OOB estimates may be less stable when the number of models is small.
3. It does not completely replace a separate test dataset for final model assessment.
4. It requires additional prediction and evaluation operations during training.

**7. Applications**

OOB Evaluation is useful in:

- Random Forest classification.
- Random Forest regression.
- Medical diagnosis.
- Fraud detection.
- Customer churn prediction.
- Predictive analytics.

**Conclusion**

Out-of-Bag Evaluation is an efficient model evaluation technique used in Bagging and Random Forest. It uses observations that are excluded from individual bootstrap samples to estimate model performance. It reduces the need for a separate validation dataset and provides an effective way to assess the generalization ability of ensemble models.

Key Terms to Remember: Out-of-Bag Evaluation, Bootstrap Sampling, Bagging, Random Forest, OOB Samples, Model Evaluation, Generalization, Validation Dataset.


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